23.XI.2016

Seminar 8

1. a) Fie $W_1 = \{X \in M_2(\mathbb{R}) \mid X\begin{pmatrix}1&0\\1&0\end{pmatrix} = O_2\}$

$W_2 = \{X \in M_2(\mathbb{R}) \mid \begin{pmatrix}1&0\\1&0\end{pmatrix} X = O_2\}$

Determinați câte o bază și dimensiunea pt. W_1, W_2, $W_1 \cap W_2$.

b) $W_1, W_2 \subseteq \mathbb{R}^3$ $\dim W_1 = \dim W_2 = 2$

Arătați că $(\exists) v \in \mathbb{R}^3$ a.î. $v \notin W_1$ și $v \notin W_2$.

2. $V = \{X \in M_2(\mathbb{R}) \mid X = X^t\}$, $A = \begin{pmatrix}0&1\\0&2\end{pmatrix}$

a) $(\forall) X \in V$, $A^t \cdot X \cdot A \in V$.

b) $f: V \longrightarrow V$, $f(X) = A^t \cdot X \cdot A$

Determinați - $M(f)$ în raport cu o bază din V.

- $\mathrm{Ker}(f)$

3. $W = \left\{\begin{pmatrix}x\\y\\z\end{pmatrix} \in \mathbb{R}^3 \mid 3x - 2y + z = 0\right\}$.

a) Aflați o bază ortonormată pentru W.

b) $\mathrm{proj}_W \begin{pmatrix}1\\1\\1\end{pmatrix} = ?$

4. $A = \begin{pmatrix}0{,}25 & 0{,}05\\0{,}75 & 0{,}95\end{pmatrix}$. $\lim_{n\to\infty} A^n = ?$

5. $q\begin{pmatrix}x\\y\end{pmatrix} = 5x^2 + 4xy + 8y^2$.

a) Scrieți matricea asociată formei pătratice A și det. dacă q e pozitiv definită, negativ definită sau nedefinită.

b) Ortodiagonalizați matricea A

$$5x^2 + 4xy + 8y^2 - 32x - 56y + 80 = 0$$

6. $A = \begin{pmatrix}1&4\\2&8\end{pmatrix}$ a) SVD

b) $\mathrm{proj}_{\mathrm{Col}(A)} \begin{pmatrix}1\\0\end{pmatrix} = ?$

c) $\begin{cases}x + 4y = 1\\2x + 8y = 0\end{cases}$ Determinați pseudosoluția sistemului de normă minimă.

1. a) $X=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$X\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}=\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}=\begin{pmatrix} a+b & 0 \\ c+d & 0 \end{pmatrix} \Rightarrow \begin{matrix} a+b=0 \\ c+d=0 \end{matrix} \Rightarrow X=\begin{pmatrix} a & -a \\ c & -c \end{pmatrix}$

$\Rightarrow W_1=\left\{\begin{pmatrix} a & -a \\ c & -c \end{pmatrix} \mid a,c\in\mathbb{R}\right\}=Sp\left\{\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix},\begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}\right\} \Rightarrow \dim_{\mathbb{R}} W_1=2$

$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}=\begin{pmatrix} a & b \\ a & b \end{pmatrix} \Rightarrow a=b=0 \Rightarrow X=\begin{pmatrix} 0 & 0 \\ c & d \end{pmatrix}$

$\Rightarrow W_2=\left\{\begin{pmatrix} 0 & 0 \\ c & d \end{pmatrix} \mid c,d\in\mathbb{R}\right\}=Sp\left\{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right\} \Rightarrow \dim_{\mathbb{R}} W_2=2$

$W_1\cap W_2=\left\{\begin{pmatrix} 0 & 0 \\ c & -c \end{pmatrix} \mid c\in\mathbb{R}\right\}=Sp\left\{\begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}\right\} \Rightarrow \dim_{\mathbb{R}} W_1\cap W_2=1$

$X\in W_1\cap W_2 \Rightarrow X\in W_1 \Rightarrow X=\begin{pmatrix} a & -a \\ c & -c \end{pmatrix}$

$X\in W_2 \Rightarrow X=\begin{pmatrix} 0 & 0 \\ c & d \end{pmatrix}$ $\Big\} \Rightarrow X=\begin{pmatrix} 0 & 0 \\ c & -c \end{pmatrix}$

b) I $W_1=W_2$

$v\notin W_1$

$W_1=Sp(v_1,v_2)$ $\Big\} \Rightarrow v\cdot v_1=v\cdot v_2=0$

II $\dim_{\mathbb{R}} W_1\cap W_2=1$

$v_1\in W_1$; $v_1\notin W_1$

$v_2\in W_2$; $v_2\notin W_2$

$w=v_1+v_2\in W_1$, $v_1\in W_1$ $\Big\} \Rightarrow \underbrace{w-v_1}_{v_2}\in W_1$ fals

$w=v_1+v_2\in W_2$, $v_2\in W_2$ $\Big\} \Rightarrow \underbrace{w-v_2}_{v_1}\in W_2$ fals

$\Big\} \Rightarrow w\notin W_1$ și $w\notin W_2$

5. a) $A=\begin{pmatrix} 5 & 2 \\ 2 & 8 \end{pmatrix}$. $\Delta_1=5>0$, $\Delta_2=\det(A)=36>0$ $\Big\} \Rightarrow q$ este pozitiv definită

a) $P_A(\lambda)=\begin{vmatrix} 5-\lambda & 2 \\ 2 & 8-\lambda \end{vmatrix}=(5-\lambda)(8-\lambda)-4=40-13\lambda+\lambda^2-4=\lambda^2-13\lambda+36=0$

$\Delta=169-144=25 \Rightarrow \lambda_{1,2}=\frac{13\pm 5}{2}$ $\begin{cases} \lambda_1=9 \\ \lambda_2=4 \end{cases}$

$V_{\lambda_1}=\left\{\begin{pmatrix} x \\ y \end{pmatrix} \mid A\begin{pmatrix} x \\ y \end{pmatrix}=\lambda_1\begin{pmatrix} x \\ y \end{pmatrix}, x,y\in\mathbb{R}\right\} \Rightarrow \begin{cases} 5x+2y=9x \Rightarrow y=2x \\ 2x+8y=9y \Rightarrow y=2x \end{cases}$ (A)

$\Rightarrow V_{\lambda_1}=\left\{\begin{pmatrix} x \\ 2x \end{pmatrix} \mid x\in\mathbb{R}\right\}=Sp\left\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\} \Rightarrow v_1=\frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

5.b) $V_{\lambda_2} = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \mid A\begin{pmatrix} x \\ y \end{pmatrix} = \lambda_2 \begin{pmatrix} x \\ y \end{pmatrix} \right\} \Rightarrow \begin{cases} 5x+2y = 4x \\ 2x+8y = 4y \end{cases} \Rightarrow x = -2y$

$\Rightarrow V_{\lambda_2} = \left\{ \begin{pmatrix} -2y \\ y \end{pmatrix} \mid y \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\} \Rightarrow u_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$A = Q \cdot D \cdot Q^t$. $Q = (u_1 \mid u_2)$

$D = \begin{pmatrix} 9 & 0 \\ 0 & 4 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} = Q \begin{pmatrix} x' \\ y' \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} x' - 2y' \\ 2x' + y' \end{pmatrix}$

$9x'^2 + 4y^2 - \frac{32}{\sqrt{5}}(x' - 2y') - \frac{56}{\sqrt{5}}(2x' + y') + 80 = 0$

$9x'^2 + 4y'^2 - \frac{144}{\sqrt{5}}x' + \frac{8y'}{\sqrt{5}}y' + 80 = 0$

$9\left(x'^2 - \frac{16}{\sqrt{5}}x' + \frac{64}{5}\right) + 4\left(y'^2 + \frac{2y'}{\sqrt{5}} + \frac{1}{5}\right) - \frac{9 \cdot 64}{5} - \frac{4}{5} + 80 = 0$

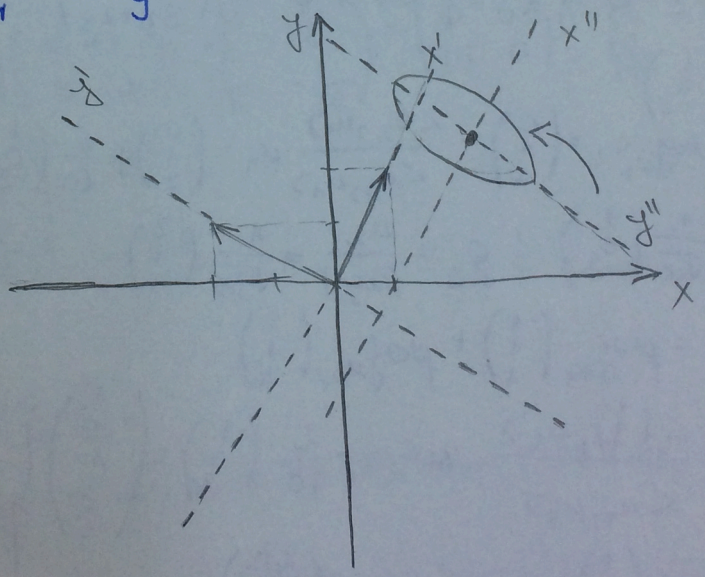
$9\left(x' - \frac{8}{\sqrt{5}}\right)^2 + 4\left(y' + \frac{1}{\sqrt{5}}\right)^2 - 36 = 0$

$\begin{cases} x'' = x' - \frac{8}{\sqrt{5}} \\ y'' = y' + \frac{1}{\sqrt{5}} \end{cases}$ $\begin{cases} x'' = 0 \Rightarrow x' = \frac{8}{\sqrt{5}} \\ y'' = 0 \Rightarrow y' = -\frac{1}{\sqrt{5}} \end{cases}$ $\Rightarrow$ noua origine este $O\left(\frac{8}{\sqrt{5}}, \frac{-1}{\sqrt{5}}\right)$ (in sistemul de axe x'Oy')

$9x''^2 + 4y''^2 = 36 \Rightarrow \frac{x''^2}{4} + \frac{y''^2}{9} = 1$ - elipsă

$q_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$q_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$



2.a) $X = X^t \Rightarrow X = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \Rightarrow V = Sp\left\{\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{x_1}, \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{x_2}, \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}_{x_3}\right\} \Rightarrow \dim_{\mathbb{R}} V = 3$

$(A^t \cdot X \cdot A)^t = A^t \cdot X^t \cdot A = A^t \cdot X \cdot A \Rightarrow A^t X A \in V$

b) $f: V \to V$

$f(X) = A^t \cdot X \cdot A = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ a+2b & b+2c \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}$

$\Rightarrow f(X) = \begin{pmatrix} 0 & 0 \\ 0 & a+4b+4c \end{pmatrix}$

$f(x_1) = f\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = x_3 = 0 \cdot x_1 + 0 \cdot x_2 + 1 \cdot x_3$

$f(x_2) = f(x_3) = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix} = 4x_3 = 0 \cdot x_1 + 0 \cdot x_2 + 4 \cdot x_3$

$\Rightarrow M(f) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 4 & 4 \end{pmatrix}$

$\operatorname{Ker} f = \left\{\begin{pmatrix} a & b \\ b & c \end{pmatrix} \mid f\begin{pmatrix} a & b \\ b & c \end{pmatrix} = 0\right\}$

$f\begin{pmatrix} a & b \\ b & c \end{pmatrix} = 0 \Rightarrow a + 4b + 4c = 0 \Rightarrow a = -4b - 4c$

$\Rightarrow \operatorname{Ker} f = \left\{\begin{pmatrix} -4b-4c & b \\ b & c \end{pmatrix} \mid b, c \in \mathbb{R}\right\} = Sp\left\{\begin{pmatrix} -4 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} -4 & 0 \\ 0 & 1 \end{pmatrix}\right\}$

3. $W = \left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid 3x - 2y + z = 0\right\}$

a) $3x - 2y + z = 0 \Rightarrow z = 2y - 3x \Rightarrow W = \left\{\begin{pmatrix} x \\ y \\ -3x+2y \end{pmatrix} \mid x, y \in \mathbb{R}\right\} = Sp\left\{\underbrace{\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}}_{v_2}\right\}$ - bază

$u_1 = v_1 = \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$

$u_2 = v_2 - \operatorname{proj}_{u_1} v_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} \cdot u_1 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \frac{6}{10}\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} -\frac{3}{5} \\ 1 \\ \frac{1}{5} \end{pmatrix}$

$q_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{10}}\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$; $q_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{35}}\begin{pmatrix} 3 \\ 1 \\ 5 \end{pmatrix}$

b) $\operatorname{proj}_W \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \operatorname{proj}_{u_1} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \operatorname{proj}_{u_2} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$\operatorname{proj}_{u_1} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{\langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, u_1 \rangle}{\langle u_1, u_1 \rangle} \cdot u_1 = -\frac{2}{10}\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} -\frac{1}{5} \\ 0 \\ \frac{3}{5} \end{pmatrix}$

$\operatorname{proj}_{u_2} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{\langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, u_2 \rangle}{\langle u_2, u_2 \rangle} \cdot u_2 = \begin{pmatrix} \frac{27}{35} \\ \frac{45}{35} \\ \frac{9}{35} \end{pmatrix}$

$\Rightarrow \operatorname{proj}_W \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{20}{35} \\ \frac{45}{35} \\ \frac{30}{35} \end{pmatrix}$

4. $A = \begin{pmatrix} 0,25 & 0,05 \\ 0,75 & 0,95 \end{pmatrix}$

$P_A(\lambda) = \begin{vmatrix} 0,25-\lambda & 0,05 \\ 0,75 & 0,95-\lambda \end{vmatrix} \overset{L_1 = L_1 + L_2}{=\!=} \begin{vmatrix} 1-\lambda & 1-\lambda \\ 0,75 & 0,95-\lambda \end{vmatrix}$

$= (1-\lambda) \begin{vmatrix} 1 & 1 \\ 0,75 & 0,95-\lambda \end{vmatrix} = (1-\lambda)(0,95-\lambda-0,75)$

$= (1-\lambda)(0,2-\lambda) = 0 \Rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 0,2 \end{cases}$

$V_{\lambda_1}: \begin{cases} 0,25x + 0,05y = x \Rightarrow 0,75x = 0,05y \Rightarrow y = 15x \\ 0,75x + 0,95y = y \end{cases}$

$\Rightarrow V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ 15x \end{pmatrix} \mid x \in \mathbb{R} \right\} = Sp\begin{pmatrix} 1 \\ 15 \end{pmatrix}$

$V_{\lambda_2}: \begin{cases} 0,25x + 0,05y = 0,2x \Rightarrow 0,05x + 0,05y = 0 \Rightarrow x = -y \Rightarrow y = -x \\ 0,75x + 0,95y = 0,2y \end{cases}$

$\Rightarrow V_{\lambda_2} = \left\{ \begin{pmatrix} x \\ -x \end{pmatrix} \mid x \in \mathbb{R} \right\} = Sp\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$A = P \cdot D \cdot P^{-1}$, unde $D = \begin{pmatrix} 1 & 0 \\ 0 & 0,2 \end{pmatrix}$, $P = \begin{pmatrix} 1 & 1 \\ 15 & -1 \end{pmatrix}$

$\Rightarrow A^n = P \cdot D^n \cdot P^{-1} = P \cdot \begin{pmatrix} 1 & 0 \\ 0 & (0,2)^n \end{pmatrix} \cdot P^{-1} \xrightarrow{n \to \infty} P \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot P^{-1}$

$P^{-1} = -\frac{1}{16}\begin{pmatrix} -1 & -1 \\ -15 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{16} & \frac{1}{16} \\ \frac{15}{16} & -\frac{1}{16} \end{pmatrix}$

$\Rightarrow A^n \xrightarrow{n \to \infty} \begin{pmatrix} 1 & 1 \\ 15 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} \frac{1}{16} & \frac{1}{16} \\ \frac{15}{16} & -\frac{1}{16} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 15 & 0 \end{pmatrix}\begin{pmatrix} \frac{1}{16} & \frac{1}{16} \\ \frac{15}{16} & -\frac{1}{16} \end{pmatrix}$

$\Rightarrow \lim_{n \to \infty} A^n = \begin{pmatrix} \frac{1}{16} & \frac{1}{16} \\ \frac{15}{16} & \frac{15}{16} \end{pmatrix}$

6. a) $A = U \cdot \Sigma \cdot V^t$

$A^t A = \begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix}\begin{pmatrix} 1 & 4 \\ 2 & 8 \end{pmatrix} = \begin{pmatrix} 5 & 20 \\ 20 & 80 \end{pmatrix}$.

$\begin{vmatrix} 5-\lambda & 20 \\ 20 & 80-\lambda \end{vmatrix} = (5-\lambda)(80-\lambda) - 400 = 400 - 85\lambda + \lambda^2 - 400 = \lambda(\lambda - 85) \Rightarrow \begin{cases} \lambda_1 = 85 \\ \lambda_2 = 0 \end{cases}$

$V_{\lambda_1}: \begin{cases} 5x + 20y = 85x \Rightarrow 80x = 20y \Rightarrow y = 4x \\ 20x + 80y = 85y \end{cases} \Rightarrow V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ 4x \end{pmatrix} \mid x \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\}$

$\Rightarrow v_1 = \frac{1}{\sqrt{17}}\begin{pmatrix} 1 \\ 4 \end{pmatrix}$

$V_{\lambda_2}: \begin{cases} 5x + 20y = 0 \Rightarrow x = -4y \\ 20x + 80y = 0 \end{cases} \Rightarrow V_{\lambda_2} = \left\{ \begin{pmatrix} -4y \\ y \end{pmatrix} \mid y \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} -4 \\ 1 \end{pmatrix} \right\} \Rightarrow v_2 = \frac{1}{\sqrt{17}}\begin{pmatrix} -4 \\ 1 \end{pmatrix}$

6. a) $V = \begin{pmatrix} \frac{1}{\sqrt{17}} & -\frac{4}{\sqrt{17}} \\ \frac{4}{\sqrt{17}} & \frac{1}{\sqrt{17}} \end{pmatrix}$ $\Sigma = \begin{pmatrix} \sqrt{85} & 0 \\ 0 & 0 \end{pmatrix}$

$u_1 = \frac{1}{\sqrt{85}} A \cdot v_1 = \frac{1}{\sqrt{85}} \begin{pmatrix} 1 & 4 \\ 2 & 8 \end{pmatrix} \frac{1}{\sqrt{17}} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \frac{1}{17\sqrt{5}} \begin{pmatrix} 17 \\ 34 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$\left\langle \begin{pmatrix} x \\ y \end{pmatrix}, u_1 \right\rangle = 0 \Rightarrow \frac{1}{\sqrt{5}}(x+2y) = 0 \Rightarrow x = -2y \Rightarrow u_2 = \begin{pmatrix} -2y \\ y \end{pmatrix} \in Sp\left\{\begin{pmatrix} -2 \\ 1 \end{pmatrix}\right\}$

$\Rightarrow u_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

b) $U = (u_1 | u_2)$

$V = (v_1 | v_2)$

$v_2 - Ker(A)$ $\quad$ $Col(A)$ (u_1)

$\perp$ $\quad$ $\perp$

$v_1 - Col(A^t)$ $\quad$ $Ker(A^t)$ (u_2)

$\Rightarrow proj_{Col(A)} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, u_1 \right\rangle}{\langle u_1, u_1 \rangle} \cdot u_1 = \frac{\frac{1}{\sqrt{5}}}{1} \begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ \frac{2}{5} \end{pmatrix}$

c) $A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$A \begin{pmatrix} x \\ y \end{pmatrix} = proj_{Col(A)} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ \frac{2}{5} \end{pmatrix} \Rightarrow \begin{cases} x + 4y = \frac{1}{5} \\ 2x + 8y = \frac{2}{5} \end{cases} \Rightarrow v = \left\{ \begin{pmatrix} \frac{1}{5} - 4y \\ y \end{pmatrix} / y \in \mathbb{R} \right\}$

$\|v\|^2 = \left(\frac{1}{5} - 4y\right)^2 + y^2 \Rightarrow y = \dots$